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Communication Protocols in Embedded Systems

Protocol concepts, categories, UART, SPI, and I²C (TWI) using ATmega32A

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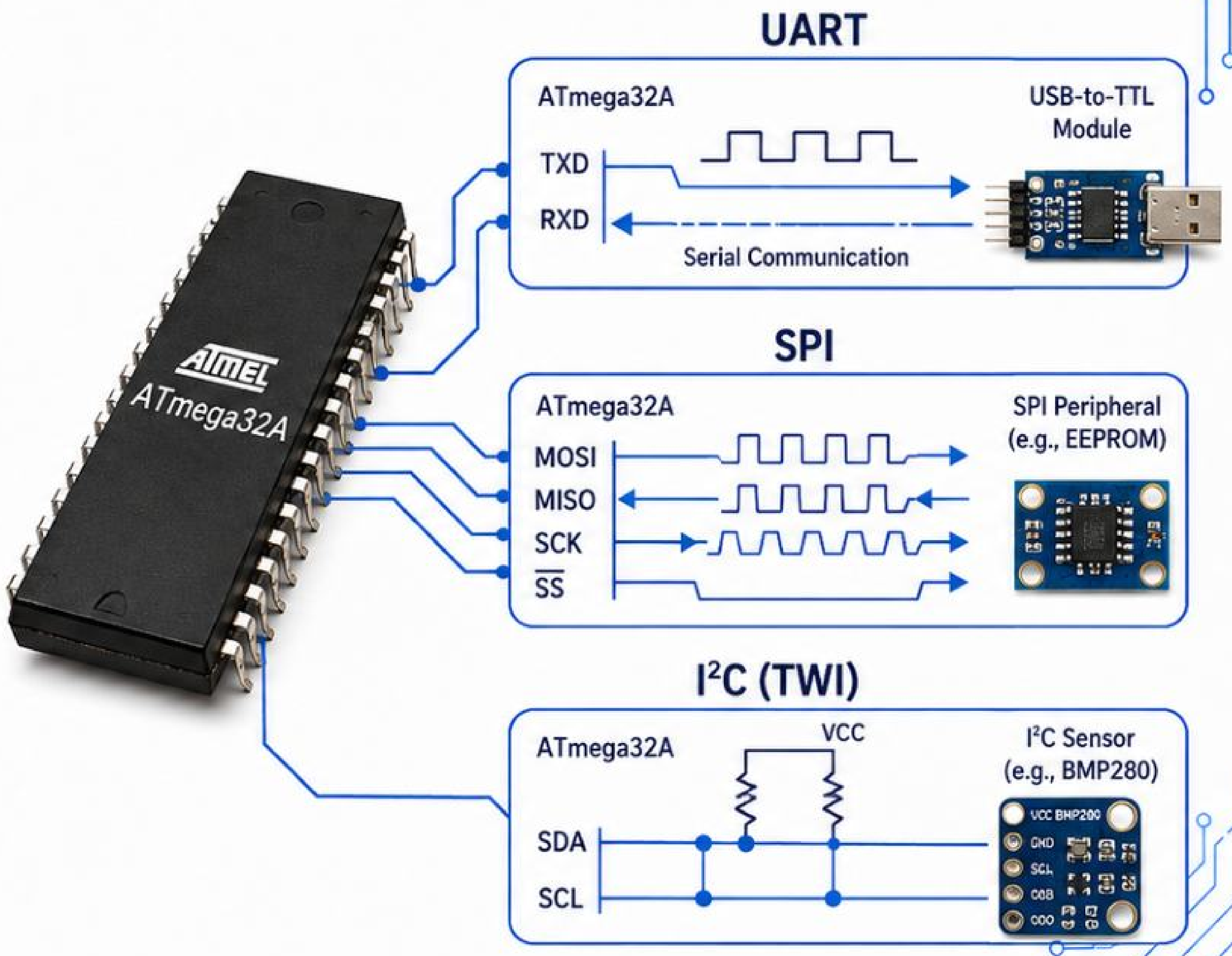

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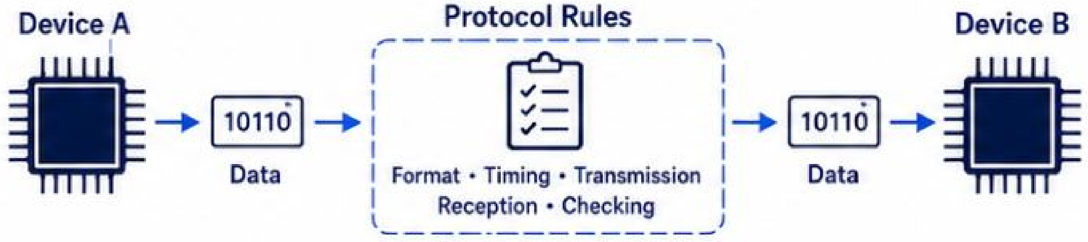
What Is a Communication Protocol?

Rules that let digital devices exchange data correctly

Definition

A protocol is a set of rules that defines how data is formatted, timed, transmitted, received, and checked between devices.

How devices communicate using protocol rules

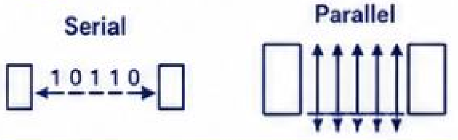


Why it matters

- Ensures data is understood correctly
- Prevents errors and miscommunication
- Enables interoperability between devices

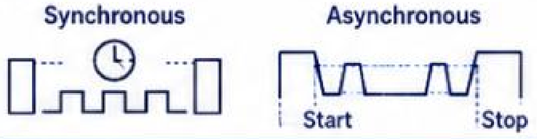
Main protocol categories

1 Serial vs Parallel



Data is sent one bit at a time or multiple bits at once.

2 Synchronous vs Asynchronous

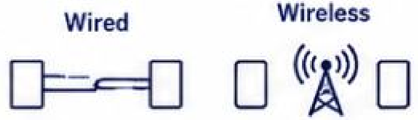


Communication uses a shared clock or start/stop timing.

3 Simplex / Half-Duplex / Full-Duplex

Data flows in one direction, both directions (not at once), or both at once.

4 Wired vs Wireless



Data is transmitted over physical cables or through the air.

In this session, we focus on wired serial protocols used with ATmega32A: UART, SPI, and I²C (TWI).

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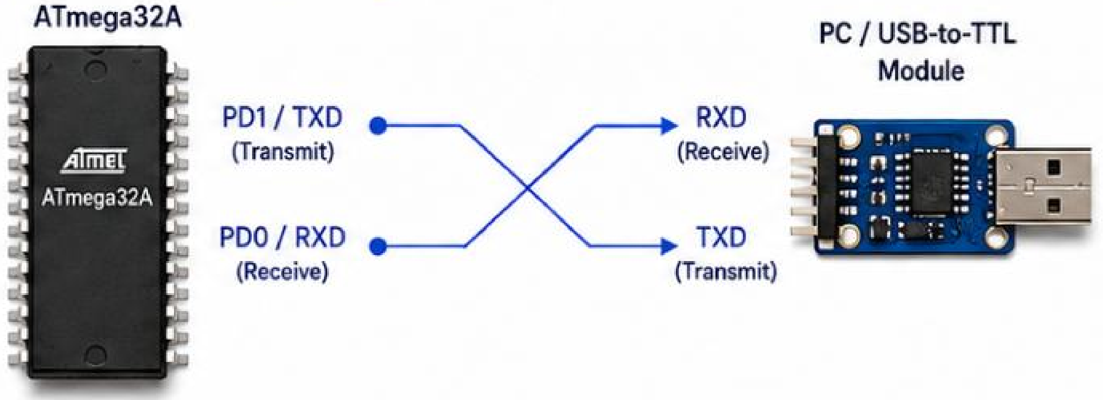
UART Communication

Universal Asynchronous Receiver/Transmitter in ATmega32A

1) UART Connection (ATmega32A ↔ PC / USB-to-TTL Module)



2) UART Frame Format

3) Asynchronous Communication

- No shared clock line.
- Both devices agree on the same baud rate.

4) Key Points

- Point-to-point communication
- Full duplex (simultaneous TX & RX)
- Simple two-wire connection
- Commonly used for serial monitor and debugging

5) ATmega32A Key Registers

UDR	UART Data Register – holds data to transmit or receive.
UCSRA	UART Control and Status Register A – status flags (RXC, TXC, UDRE, FE, DOR, PE).
UCSRB	UART Control and Status Register B – enables TX/RX and interrupts.
UCSRC	UART Control and Status Register C – frame format: data bits, parity, stop bits.
UBRRH / UBRRL	UART Baud Rate Register (High / Low) – sets the baud rate.

6) Common UART Pins on ATmega32A

RXD = PD0
TXD = PD1

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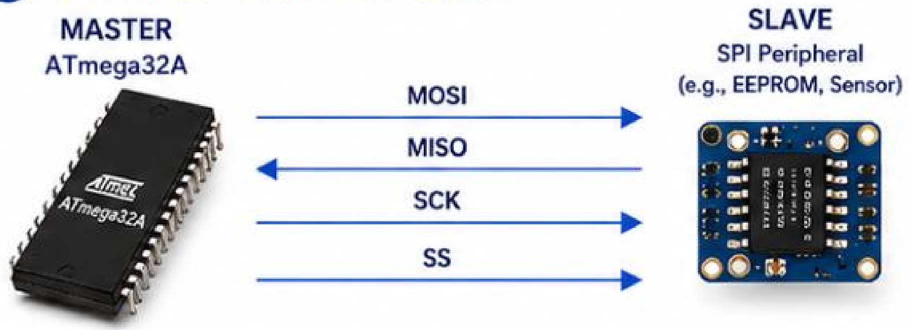
SPI Communication

Serial Peripheral Interface in ATmega32A

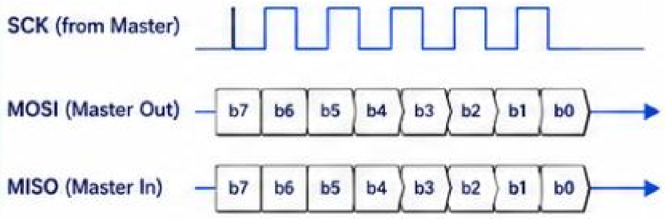
1 SPI Master–Slave Connection



2 What is SPI?

- Synchronous serial protocol
- Master provides the clock (SCK)
- Fast, full-duplex communication between master and slave devices
- Simple 4-wire interface

3 Timing Concept



Data is sampled on SCK edges as defined by CPOL/CPHA.

4 Key Points

- High speed communication
- Short distance (on-board)
- One master can control multiple slaves using separate $\overline{SS}$ lines
- Full-duplex (transmit and receive simultaneously)

5 ATmega32A Pins (SPI)

MOSI	PB5
MISO	PB6
SCK	PB7
$\overline{SS}$	PB4

6 ATmega32A Key Registers

SPCR	SPI Control Register Configures SPI enable, master/slave, clock rate, and mode.
SPSR	SPI Status Register Contains status flags (SPIF, WCOL, etc.).
SPDR	SPI Data Register Read/write data for SPI transmission and reception.

Why This Matters

- Enables high-speed data exchange with peripherals like EEPROMs, sensors, ADCs, LCD drivers, etc.
- Widely used in embedded systems for efficient communication.

CPOL (Clock Polarity) and CPHA (Clock Phase) bits in SPCR define the SPI mode (0 to 3).

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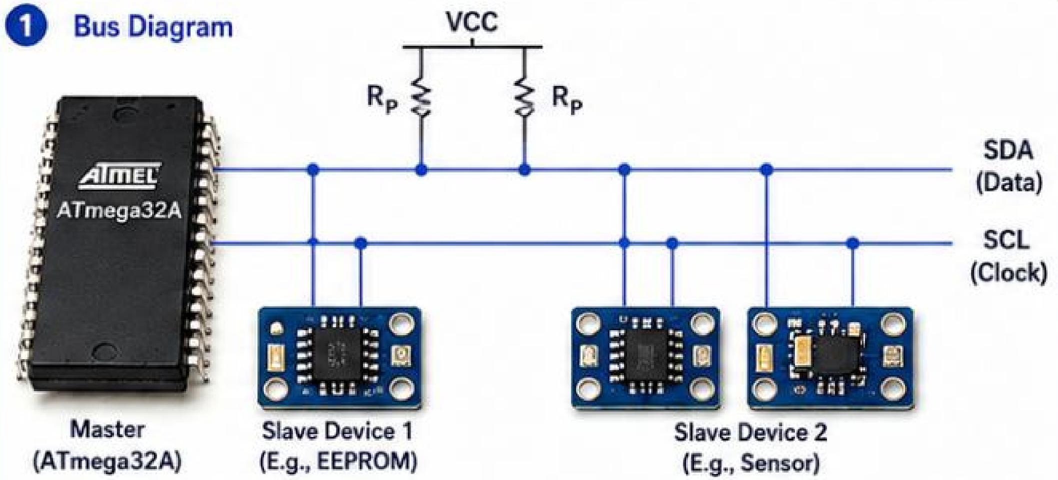
I²C / TWI Communication

Two-wire serial bus used in ATmega32A

1 Bus Diagram



2 What is I²C / TWI?

- Synchronous serial bus
- Uses only two wires: SDA (data) and SCL (clock)
- Address-based communication between master and slave devices
- Supports multiple devices on the same bus

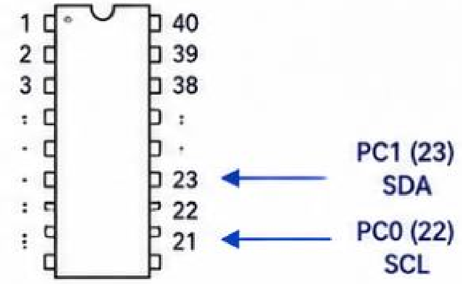
3 Protocol Sequence

START (S) → ADDRESS (7/10-bit) → ACK (A) → DATA (8-bit) → ACK (A) → STOP (P)

4 Key Points

- Only two signal lines (SDA and SCL)
- Half duplex communication
- Multiple devices can share the same bus
- Open-drain / open-collector style outputs require pull-up resistors to VCC

5 ATmega32A Pins



6 ATmega32A Key Registers

Register	Description
TWBR	Bit rate register. Sets the SCL clock frequency.
TWSR	Status register and prescaler. Contains status codes and prescaler bits (TWPS1:0).
TWAR	(Slave) Address register. Holds the 7-bit slave address.
TWDR	Data register. Holds data to transmit or receive.
TWCR	Control register. Enables I²C, generates START/STOP, acknowledges, and enables interrupts.

7 Note

Microchip calls it TWI, but it is commonly known as I²C.

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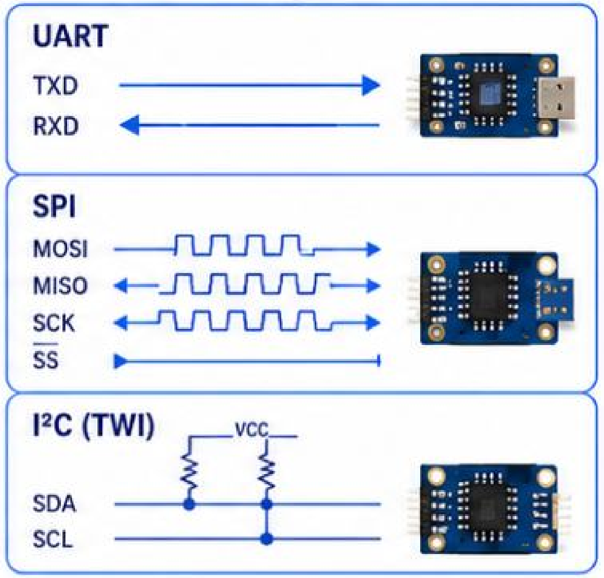
UART vs SPI vs I²C

Comparison of the main wired serial protocols

Feature	UART	SPI	I²C (TWI)
Number of signal lines	2 (TXD, RXD)	4 basic (MOSI, MISO, SCK, SS)	2 (SDA, SCL)
Clock type	None (asynchronous)	Master-generated clock	Clocked (shared clock line)
Duplex type	Full duplex	Full duplex	Half duplex
Addressing	None (point-to-point)	None (chip select per device)	Yes (7-bit / 10-bit addresses)
Typical topology	Point-to-point	Star (master to multiple slaves)	Multi-drop bus
Speed (qualitative)	Medium	High	Low to Medium
Main advantage	Simple, minimal wiring	Very fast, high throughput	Few wires, many devices
Main limitation	No addressing, limited speed	More wires, no built-in addressing	Lower speed, bus capacitance limits
Typical applications	Debug consoles, GPS modules, Bluetooth modules	Displays, sensors, memories, ADC/DAC	Sensors, EEPROM, RTC, expanders

Protocol Signal Overview



Decision Guide

- Choose UART for simple serial links and debugging.
- Choose SPI for high-speed peripherals.
- Choose I²C for many low-speed devices on one bus.

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ATmega32A Communication Registers

Simplified register guide for UART, SPI, and TWI

UART

UDR	Data buffer
UCSRA	Status flags
UCSRB	Enable TX/RX and interrupts
UCSRC	Frame format
UBRRH/UBRRL	Baud rate

SPI

SPCR	Control
SPSR	Status
SPDR	Data

TWI

TWBR	Bit rate
TWSR	Status / prescaler
TWAR	Slave address
TWDR	Data
TWCR	Control

These are memory-mapped peripheral registers used to configure communication, start transfers, and monitor status flags.

ATmega32A – PERIPHERALS REGISTERS SUMMARY

1. DIGITAL I/O PORTS

Register	Function / What it does
DDRA, DDRB, DDRC, DDRD	Data Direction Register – 1 = Output, 0 = Input
PORTA, PORTB, PORTC, PORTD	Output value register / enables pull-up when bit = 1 (in input mode)
PINA, PINB, PINC, PIND	Input Pins Register – reads the logic level on pins
PINx (Writing 1)	Toggles the corresponding output bit (if pin is output)

2. EXTERNAL INTERRUPTS

Register	Function / What it does
MCUCR	Interrupt Sense Control – Selects triggering conditions for INT0, INT1
MCUCSR	Interrupt Control/Status – ISC2 for INT2, flags for interrupts
GIFR	General Interrupt Flag Register – Interrupt flags (INTF0, INTF1, INTF2)
GICR	General Interrupt Control Register – Enable/Disable external interrupts

3. TIMERS / COUNTERS

Register		Function / What it does
Timer0 (8-bit)	TCCR0	Timer/Counter Control Register – Clock source, mode, prescaler
	TCNT0	Timer/Counter Register – Counts timer value
	OCR0	Output Compare Register – Compare value for match
Timer1 (16-bit)	TCCR1A	Control Register A – Waveform mode, compare output mode
	TCCR1B	Control Register B – Clock source, prescaler, input capture edge
	TCNT1H / TCNT1L	16-bit Timer/Counter Register
	OCR1AH / OCR1AL	Output Compare Register A (16-bit)
	OCR1BH / OCR1BL	Output Compare Register B (16-bit)
	ICR1H / ICR1L	Input Capture Register (16-bit)
Timer2 (8-bit)	TCCR2	Timer/Counter Control Register – Clock source, mode, prescaler
	TCNT2	Timer/Counter Register
	OCR2	Output Compare Register

4. ANALOG TO DIGITAL CONVERTER (ADC)

Register	Function / What it does
ADMUX	Multiplexer Selection Register – Selects ADC channel and reference voltage
ADCSRA	ADC Control and Status Register A – Enable ADC, start conversion, prescaler, flags
ADCH	ADC Data Register High Byte
ADCL	ADC Data Register Low Byte
SFIOR	Special Function IO Register – ADTS bits for Auto Trigger Source
DIDR0	Digital Input Disable Register – Disables digital input on ADC pins (reduces power)

5. ANALOG COMPARATOR

Register	Function / What it does
ACSR	Analog Comparator Control and Status Register – Enable comparator, output state, interrupt flag
ADMUX	(Used with Comparator) Selects comparator negative input (AIN1)
SFIOR	Analog Comparator Multiplexer Enable – Selects AIN0 pin source
GICR	Enable Comparator Interrupt (ACI)
GIFR	Comparator Interrupt Flag (ACIF)

6. USART

Register	Function / What it does
UBRRH / UBRRL	USART Baud Rate Register – Sets baud rate
UCSRA	USART Control and Status Register A – Status flags, double speed mode
UCSRB	USART Control and Status Register B – Enable TX, RX, interrupts
UCSRC	USART Control and Status Register C – Frame format, stop bits, parity
UDR	USART Data Register – Transmit / Receive data buffer

7. SPI

Register	Function / What it does
SPCR	SPI Control Register – Enable SPI, master/slave, clock polarity, phase
SPSR	SPI Status Register – SPIF flag, write collision flag, SPI2X
SPDR	SPI Data Register – Transmit / Receive data buffer

8. TWI (I²C)

Register	Function / What it does
TWBR	TWI Bit Rate Register – Sets SCL clock frequency
TWSR	TWI Status Register – Prescaler bits and status codes
TWAR	TWI Address Register – Slave address
TWDR	TWI Data Register – Transmit / Receive data buffer
TWCR	TWI Control Register – Enable TWI, start, stop, ACK, interrupt

9. WATCHDOG TIMER

Register	Function / What it does
WDTCR	Watchdog Timer Control Register – Enable WDT, timeout period, reset
MCUSR	MCU Status Register – WDRF flag shows reset caused by WDT

10. EEPROM

Register	Function / What it does
EEARH / EEARL	EEPROM Address Register – Selects EEPROM address
EEDR	EEPROM Data Register – Data to be read or written
EECR	EEPROM Control Register – Read/Write enable, interrupt enable
EECR (EERIE)	EEPROM Ready Interrupt Enable
EECR (EEPE)	EEPROM Write Enable

11. JTAG (Boundary Scan)

Register	Function / What it does
MCUCSR	JTAG Enable / Disable
MCUCR	JTAG Enable (JTD bit) Disables JTAG for normal I/O use

12. SYSTEM CONTROL

Register	Function / What it does
MCUCR	MCU Control Register – Reset, sleep modes, power control
MCUSR	MCU Status Register – Reset flags (PORF, EXTRF, WDRF, BORF)
SFIOR	Special Function IO Register – Various system options
SMCR	Sleep Mode Control Register – Select sleep mode

Note: All registers are memory-mapped I/O registers. Their addresses are defined in the ATmega32A datasheet.

SUMMARY – WHAT WE COVERED

1 ELECTRONICS SWITCHES

- Types: Push Button, Toggle, Slide, DIP, Rotary
- SPST, SPDT, Normally Open (NO), Normally Closed (NC)
- Pull-up / Pull-down Resistors
- Debouncing (Hardware / Software)
- Applications Examples

2 TOOLCHAIN & BUILD PROCESS

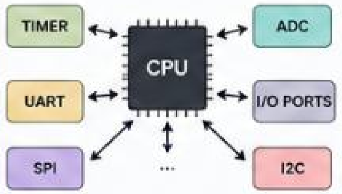
- Write Code (C/C++)
- Preprocessing
- Compilation → Object File (.o)
- Assembly
- Linking → Executable (.elf)
- Convert to Hex (.hex)
- Program to MCU

3 ECLIPSE PROJECT STRUCTURE

Includes	Header files
src	Source files (.c)
include	User headers (.h)
Debug	Build output files
Makefile	Build configuration

4 WHAT IS A PERIPHERAL?

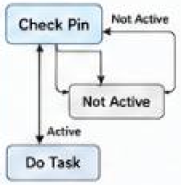
- Hardware module inside the MCU that performs a specific task.
- Works with CPU through Registers.
- Examples: Timers, ADC, UART, SPI, I2C, etc.



5 POLLING vs INTERRUPTS

Polling

- CPU continuously checks the pin.
- Simple but wastes CPU time.



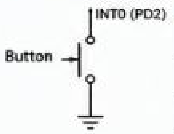
Interrupts

- MCU does other work and reacts when event occurs.
- Faster response, efficient.

6 EXTERNAL INTERRUPTS

- ATmega32 has 3 external interrupts: INT0, INT1, INT2.
- Triggered by: Low level, Any logical change, Falling edge, Rising edge.
- Registers: MCUCR, GICR, GIFR.
- Used for fast response to external events.

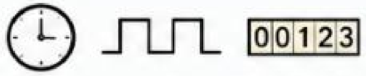
Example Circuit



Interrupt	Vector Address
INT0	0x0002
INT1	0x0004
INT2	0x0006

7 TIMERS & COUNTERS

- 3 Timers: Timer0 (8-bit), Timer1 (16-bit), Timer2 (8-bit).
- Modes: Normal, CTC, Fast PWM, Phase Correct PWM.
- Uses: Time delay, Event counting, PWM generation, Input capture.
- Registers: TCCRnA, TCCRnB, TCNTn, OCRn, TIMSK, TIFR.



8 TIMER CALCULATIONS (EXAMPLE)

General Formula:

$$T = \frac{N \times Prescaler}{F_{CPU}}$$

Where:
T = Time (s)
N = Number of counts
F_{CPU} = CPU Frequency

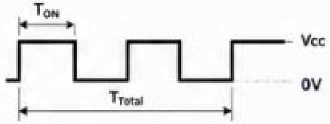
Example:
F_{CPU} = 16 MHz, Prescaler = 256
Timer0 (8-bit), N = 256

T = (256 × 256) / 16,000,000 = 4.096 ms

9 PWM GENERATION

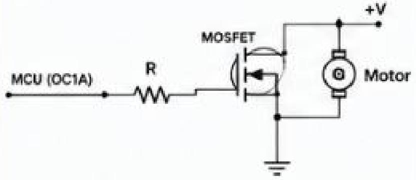
- PWM = Pulse Width Modulation.
- Generated using Timer in PWM mode.
- Key terms: Frequency, Period, Duty Cycle.

$$\text{Duty Cycle (\%)} = \frac{T_{ON}}{T_{Total}} \times 100\%$$



10 PRACTICAL ACTIVITY

- LED Brightness Control using PWM.
- DC Motor Speed Control using PWM.
- Components: MCU, MOSFET/Driver, Motor/LED.
- Students implement code, observe output and modify duty cycle.



PERIPHERALS SUMMARY – ATmega32

I/O PORTS	INTERRUPTS	TIMERS	ANALOG	COMMUNICATION	MEMORY & OTHERS	SYSTEM
• PORTA, PORTB, PORTC, PORTD • DDRx, PORTx, PINx • Digital Input/Output • Pull-up Resistors	• INT0, INT1, INT2 • MCUCR, MCUCSR • GICR, GIFR • External Interrupt Vectors	• Timer0 (8-bit) • Timer1 (16-bit) • Timer2 (8-bit) • TCCRnA, TCCRnB • TCNTn, OCRn • TIMSK, TIFR	• ADC (10-bit, 8-ch) • ADCSRA, ADMUX, ADCH, ADCL, DIDR0 • Analog Comparator • ACSR, ADMUX SFIOR (ACME)	• USART (UBRRH, UBRRL, UCSRA, UCSRB, UCSRC, UDR) • SPI (SPCR, SPSR, SPDR) • TWI (I2C) (TWBR, TWSR, TWAR, TWDR, TWCR)	• EEPROM (EEAR, EEDR, EECR) • Watchdog Timer (WDTCR, MCUSR) • JTAG (MCUCSR, MCUCR)	• MCUCR (Clock, Reset) • MCUSR (Status Flags) • SMCR (Sleep Control) • SFIOR (Special Features)

Key Takeaway: Peripherals extend the power of the MCU. Understanding their registers and operation is key to embedded system design.